

Answers To: SQL Practical 4

Question 1

```
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6
7  -- 1. View the first 100 rows of the dataset to understand its structure.
8
9  SELECT TOP 100*
10 FROM [Soccer_Analysis_Db1].[dbo].[ketros_sa_soccer_dataset_advanced]
11
12
```

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Results Messages

	player_name	team	date_of_birth	age	marital_status	number_of_kids	nationality	country_of_birth	position	preferred_foot
1	Siyanda Dlamini	Stellenbosch FC	1995-05-28	30	Widowed	0	South African	South African	Defender	Right
2	Thabo Ndlovu	Cape Town City	2004-06-01	21	Single	1	Zimbabwean	Zimbabwean	Forward	Right
3	Vusi Molefe	Stellenbosch FC	2006-09-20	19	Single	0	Nigerian	Nigerian	Forward	Left
4	Thembi Mahlangu	Bloemfontein Celtic	2004-02-08	21	Divorced	0	Zambian	Zambian	Goalkeeper	Left
5	Nokuthula Sithole	Polokwane City	2003-03-18	22	Divorced	3	Nigerian	Nigerian	Goalkeeper	Both
6	Thembi Sithole	Kaizer Chiefs	1989-02-14	36	Married	2	Nigerian	Nigerian	Goalkeeper	Left
7	Siyanda Mahlangu	Chippa United	1989-12-09	36	Single	0	Zambian	Zambian	Defender	Both
8	Lerato Mashaba	Polokwane City	1998-01-31	27	Widowed	0	Ghanaian	Ghanaian	Forward	Right
9	Nomsa Mahlangu	Polokwane City	1991-07-16	34	Divorced	0	Malawian	Malawian	Forward	Left
10	Tumelo Khumalo	Kaizer Chiefs	1996-08-28	29	Married	4	Malawian	Malawian	Forward	Both
11	Gugu Molefe	TS Galaxy	2007-01-29	18	Widowed	0	South African	South African	Defender	Left
12	Thembi Mokoena	Stellenbosch FC	1995-07-03	30	Widowed	0	Zimbabwean	Zimbabwean	Goalkeeper	Right
13	Thabo Sithole	Bloemfontein Celtic	2000-12-17	25	Single	0	Zimbabwean	Zimbabwean	Midfielder	Both
14	Gugu Mabena	SuperSport United	2000-09-19	25	Widowed	0	Zambian	Zambian	Goalkeeper	Right
15	Kagiso Phiri	Bloemfontein Celtic	1990-06-30	35	Divorced	0	Zambian	Zambian	Midfielder	Both
16	Thembi Tehahalala	Cape Town City	2003-01-22	22	Married	4	Ghanaian	Ghanaian	Defender	Both

Query executed successfully. DESKTOP-I4EJ2HG

Question 2

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11
12  --2. Count the total number of players in the dataset.
13
14  SELECT COUNT(*) AS Total_Number_Of_Players
15  FROM [Soccer_Analysis_Db1].[dbo].[ketros_sa_soccer_dataset_advanced]
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Results Messages

	Total_Number_Of_Players
1	300

Question 3

SQLQuery1.sql...G\User (68))*

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17  -----
18  --3.List all unique teams in the league.
19  -----
20  SELECT DISTINCT team AS Team_Names
21  FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
22
23
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```

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Results Messages

	Team_Names
1	AmaZulu FC
2	Bloemfontein Celtic
3	Cape Town City
4	Chippa United
5	Golden Arrows
6	Kaizer Chiefs
7	Mamelodi Sundo...
8	Moroka Swallows
9	Orlando Pirates
10	Polokwane City
11	Richards Bay FC
12	Royal AM
13	Sekhukhune Unit...
14	Stellenbosch FC
15	SuperSport United
16	TS Galaxy

Question 4

SQLQuery1.sql...G\User (68))*

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23  -----
24  --4. Count how many players are in each team.
25  -----
26  SELECT team,
27         COUNT(*) AS Number_Of_Players
28  FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
29  GROUP BY team;
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Results Messages

	team	Number_Of_Players
8	Moroka S...	19
9	Orlando Pi...	9
10	Polokwan...	27
11	Richards ...	13
12	Royal AM	18
13	Sekhukhu...	17
14	Stellenbos...	20
15	SuperSpor...	20
16	TS Galaxy	20

Query executed successfully.

Question 5

30 --5. Identify the top 10 players with the most goals.
31
32
33 SELECT TOP 10*
34
35 FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
36 ORDER BY goals DESC;

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Results Messages

	player_name	team	date_of_birth	age	marital_status	number_of_kids	nationality	country_of_birth	position	preferred_foot	height_cm	weight_kg	jersey_number	injury_status	ag
1	Siyanda Mabena	Sekhukhune United	1992-12-15	33	Widowed	0	Mozambican	Mozambican	Midfielder	Right	169	76	31	Recovering	Sc
2	Kagiso Phiri	Richards Bay FC	1995-06-16	30	Divorced	3	Ghanaian	Ghanaian	Midfielder	Both	165	74	41	Injured	Sc
3	Zanele Phiri	TS Galaxy	1993-09-12	32	Divorced	2	Zimbabwean	Zimbabwean	Forward	Left	177	91	89	Healthy	Nk
4	Zanele Molefe	SuperSport United	1999-08-13	26	Divorced	3	South African	South African	Midfielder	Left	183	65	5	Recovering	S/
5	Nokuthula Mabena	SuperSport United	1998-02-03	27	Single	0	Malawian	Malawian	Midfielder	Right	170	62	56	Injured	Pl
6	Lindiwe Molefe	Chippa United	1991-08-31	34	Widowed	1	Ghanaian	Ghanaian	Midfielder	Both	184	75	55	Healthy	Sc
7	Nokuthula Mthembu	Mamelodi Sundowns	1994-11-27	31	Single	0	Malawian	Malawian	Midfielder	Both	181	68	83	Healthy	Nk
8	Karabo Phiri	Chippa United	2002-03-08	23	Married	0	Nigerian	Nigerian	Forward	Right	185	89	82	Healthy	Sc
9	Lindiwe Radebe	Chippa United	1996-02-14	29	Single	1	Zimbabwean	Zimbabwean	Forward	Right	181	74	40	Injured	Nk
10	Vusi Mabena	Richards Bay FC	2001-06-19	24	Widowed	1	Malawian	Malawian	Midfielder	Both	168	82	20	Injured	Pr

Question 6

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--6. Find the average salary for players in each team.

SELECT team,

AVG(average_salary_zar) As Average_Salary

FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]

GROUP BY team;

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Results

Messages

	team	Average_Salary
1	AmaZulu FC	172467.567826705
2	Bloemfontein Celtic	163683.839154412
3	Cape Town City	208407.431770833
4	Chippa United	180947.63374256
5	Golden Arrows	199057.643851902
6	Kaizer Chiefs	188954.461621094
7	Mamelodi Sundowns	194404.484375
8	Moroka Swallows	186786.828330592
9	Orlando Pirates	178570.592881944
10	Polokwane City	171738.877314815
11	Richards Bay FC	193006.200270433
12	Royal AM	209078.330729167
13	Sekhukhune United	169945.162683824
14	Stellenbosch FC	182326.734960937
15	SuperSport United	195482.980664062
16	TS Galaxy	184719.574609375

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Query executed successfully.

Question 7

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46 --7. Retrieve the top 10 players with the highest market value.
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SELECT TOP 10 *
FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
ORDER BY market_value_zar DESC ;

```

100 % No issues found

Results Messages

	player_name	team	date_of_birth	age	marital_status	number_of_kids	nationality	country_of_birth	position	preferred_foot	height
1	Ayanda Mabaso	Bloemfontein Celtic	1989-07-03	36	Widowed	0	South African	South African	Goalkeeper	Left	189
2	Lebogang Mabena	Mamelodi Sundowns	1993-01-06	32	Divorced	0	Ghanaian	Ghanaian	Midfielder	Right	180
3	Gugu Mashaba	Moroka Swallows	2000-12-10	25	Divorced	2	South African	South African	Goalkeeper	Right	171
4	Kagiso Mokoena	Golden Arrows	2003-06-11	22	Single	0	Zambian	Zambian	Defender	Left	176
5	Boitumelo Radebe	Sekhukhune United	2002-09-24	23	Single	0	Ghanaian	Ghanaian	Goalkeeper	Right	171
6	Thabo Ndlovu	Mamelodi Sundowns	1991-06-16	34	Single	0	Nigerian	Nigerian	Midfielder	Both	183
7	Sipho Mashaba	Golden Arrows	1988-02-12	37	Married	4	Zambian	Zambian	Midfielder	Right	171
8	Lerato Ngobeni	AmaZulu FC	1992-08-18	33	Single	1	Malawian	Malawian	Goalkeeper	Right	189
9	Thabo Tshabalala	Cape Town City	1992-05-05	33	Single	1	Mozambican	Mozambican	Defender	Right	192
10	Lerato Tshabalala	Chippa United	1995-01-08	30	Widowed	0	Zambian	Zambian	Defender	Right	192

Question 8

SQLQuery1.sql...G\User (68)*

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```

SELECT TOP 10 *
FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
ORDER BY market_value_zar DESC ;

--8. Calculate the average passing accuracy for each position.

SELECT position,
       AVG(passing_accuracy) AS Average_Pass_Accuracy
FROM [Soccer_Analysis_Db1].[dbo].[ketrosoccerdataset_advanced]
GROUP BY position;

```

100 % No issues found

Results Messages

	position	Average_Pass_Accuracy
1	Defender	82.6642859322684
2	Forward	83.210843373494
3	Goalke...	83.9397055682014
4	Midfielder	82.817721596247

Question 9

SQLQuery1.sql...G\User (68))*

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60
61 --9. Compare shot accuracy with goals to find correlations.
62
63 SELECT position,
64        AVG(shot_accuracy) AS Average_Shot_Accu,
65        AVG(goals) AS Average_Goals
66 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
67 GROUP BY position;

```

100 % No issues found

Results Messages

	position	Average_Shot_Accu	Average_Goals
1	Defender	67.1785712105887	4
2	Forward	64.2204823321607	53
3	Goalkeeper	62.9911766613231	5
4	Midfielder	66.2227848873863	61

Question 10

SQLQuery1.sql...G\User (68))*

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69
70 --10. Compute total goals and assists for each team.
71
72 SELECT team,
73        SUM(goals) AS Number_Of_Goals,
74        SUM(assists) AS Number_of_Assists
75 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
76 GROUP BY team;

```

100 % No issues found

Results Messages

	team	Number_Of_Goals	Number_of_Assists
1	AmaZulu FC	546	568
2	Bloemfontein Celtic	613	566
3	Cape Town City	374	284
4	Chippa United	684	205
5	Golden Arrows	676	519
6	Kaizer Chiefs	597	481
7	Mamelodi Sundowns	570	475
8	Moroka Swallows	729	452
9	Orlando Pirates	170	225
10	Polokwane City	1300	783
11	Richards Bay FC	453	273
12	Royal AM	544	480
13	Sekhukhune United	842	608
14	Stellenbosch FC	725	514
15	SuperSport United	580	514
16	TS Galaxy	541	337

Question 11

SQLQuery1.sql...G\User (68))

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79 --11. Count players by their marital status.
80
81 SELECT marital_status,
82        COUNT(*) AS Number_Of_Players
83 FROM [Soccer_Analysis_Db1].[dbo].[ketro_sa_soccer_dataset_advanced]
84 GROUP BY marital_status;
85

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Results Messages

	marital_status	Number_Of_Players
1	Divorced	78
2	Married	65
3	Single	79
4	Widowed	78

Question 12

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88 --12. Count players by nationality.
89
90 SELECT nationality,
91        COUNT(*) AS Number_Of_Players
92
93 FROM [Soccer_Analysis_Db1].[dbo].[ketro_sa_soccer_dataset_advanced]
94 GROUP BY nationality
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Results Messages

	nationality	Number_Of_Players
1	Ghanaian	47
2	Malawian	42
3	Mozambican	37
4	Nigerian	39
5	South African	46
6	Zambian	44
7	Zimbabwean	45

Question 13

SQLQuery1.sql...G\User (68))*

```

96
97 --13. Find average market value grouped by nationality.
98
99 SELECT nationality,
100         AVG(market_value_zar) AS Average_Market_Value_ZAR
101
102 FROM [Soccer_Analysis_Db1].[dbo].[ketro_sa_soccer_dataset_advanced]
103 GROUP BY nationality
104
105

```

100 % No issues found

Results Messages

	nationality	Average_Market_Value_ZAR
1	Ghanaian	12298308.0079787
2	Malawian	11727296.014881
3	Mozambican	14736203.4746622
4	Nigerian	15069261.3477564
5	South African	12037870.1983696
6	Zambian	12777043.0852273
7	Zimbabwean	10367909.4708333

Question 14

SQLQuery1.sql...G\User (68))*

```

104
105 --14. Determine how many player contracts end in each year.
106
107 SELECT contract_end_year,
108         COUNT(*) AS Number_Of_Players
109 FROM [Soccer_Analysis_Db1].[dbo].[ketro_sa_soccer_dataset_advanced]
110 GROUP BY contract_end_year

```

100 % No issues found

Results Messages

	contract_end_year	Number_Of_Players
1	2026	63
2	2027	70
3	2028	52
4	2029	50
5	2030	65

Question 15

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112
113 --15. Identify players whose contracts end next year.
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```

SELECT contract_end_year,
       COUNT(*) AS Number_Of_Players
FROM   [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
WHERE  contract_end_year = 2026
GROUP BY contract_end_year

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Results Messages

	contract_end_year	Number_Of_Players
1	2026	63

Question 16

SQLQuery1.sql...G\User (68)) * 7 X

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--16. Summarize the number of players by injury status.

SELECT injury_status,
       COUNT(*) AS Number_Of_Players
FROM   [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
GROUP BY injury_status

```

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Results Messages

	injury_status	Number_Of_Players
1	Healthy	99
2	Injured	97
3	Recovering	104

Question 17

SQLQuery1.sql...G\User (68))*

```

128
129 --17. Calculate goals per match ratio for each player
130
131 SELECT player_name,
132         team,
133         date_of_birth,
134         age,
135         goals/matches_played AS Goals_Per_Match_Ratio
136 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
137 ORDER BY Goals_Per_Match_Ratio DESC
138

```

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Results Messages

	player_name	team	date_of_birth	age	Goals_Per_Match_Ratio
1	Thembi Zulu	Sekhukhune United	2003-01-10	22	24
2	Thabo Mthembu	Moroka Swallows	1999-07-27	26	17
3	Gugu Tshabalala	TS Galaxy	1993-04-05	32	6
4	Lindiwe Mabaso	Mamelodi Sundowns	1997-04-25	28	5
5	Mpho Zulu	TS Galaxy	2007-01-03	18	5
6	Sipho Tshabalala	Stellenbosch FC	2004-11-10	21	4
7	Lindiwe Zulu	Kaizer Chiefs	1993-11-18	32	3
8	Karabo Tshabal...	Stellenbosch FC	1991-02-18	34	3
9	Boitumelo Nkosi	Stellenbosch FC	2000-04-23	25	3
10	Sipho Phiri	Royal AM	1995-01-13	30	2
11	Thembi Mokoena	Sekhukhune United	1994-12-23	31	1
12	Thabo Ndlovu	Mamelodi Sundowns	1991-06-16	34	1
13	Ayanda Sithole	Golden Arrows	1993-07-22	32	1
14	Kagiso Phiri	Richards Bay FC	1995-06-16	30	1
15	Gugu Dlamini	Moroka Swallows	1988-05-09	37	1
16	Thabo Sithole	TS Galaxy	2003-12-24	22	1

Query executed successfully.

Question 18

SQLQuery1.sql...G\User (68)*

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137 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
138 ORDER BY Goals_Per_Match_Ratio DESC
139
140 --18. Count how many players are managed by each agent.
141
142 SELECT agent,
143        COUNT(*) AS Number_Of_Players
144 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
145 GROUP BY agent
146

```

100 % No issues found

Results Messages

	agent	Number_Of_Players
1	None	62
2	PlayerFirst	63
3	ProSport	62
4	SA Elite Agents	51
5	SoccerLink A...	62

Question 19

SQLQuery1.sql...G\User (68)* SQLQuery2.sql - not connected

```

147 --19. Calculate average height and weight by player position.
148
149 SELECT position,
150        AVG(weight_kg) AS Average_Height_Of_Players,
151        AVG(height_cm) AS Average_Weight_Of_Players
152 FROM [Soccer_Analysis_Db1].[dbo].[ketros_soccer_dataset_advanced]
153 GROUP BY position
154

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100 % No issues found

Results Messages

	position	Average_Height_Of_Players	Average_Weight_Of_Players
1	Defender	78	182
2	Forward	78	179
3	Goalkeeper	78	179
4	Midfielder	76	179

Question 20

SQLQuery1.sql...G\User (68)) * SQLQuery2.sql - not connected

```

154
155
156 --20. Identify players with the highest combined goals and assists.
157
158 SELECT player_name,
159         team,
160         date_of_birth,
161         age,
162         goals+assists AS Combined_Goals_and_Assists
163 FROM     [Soccer_Analysis_Db1].[dbo].[ketros_sa_soccer_dataset_advanced]
164
165 ORDER by Combined_Goals_and_Assists DESC

```

100 % No issues found

Results Messages

	player_name	team	date_of_birth	age	Combined_Goals_and_Assists
1	Siyanda Mabena	Sekhukhune United	1992-12-15	33	193
2	Zanele Molefe	SuperSport United	1999-08-13	26	192
3	Vusi Radebe	Stellenbosch FC	2000-03-22	25	188
4	Ayanda Sithole	Golden Arrows	1993-07-22	32	179
5	Ayanda Sithole	Bloemfontein Celtic	2004-11-16	21	178
6	Vusi Molefe	Stellenbosch FC	2006-09-20	19	176
7	Thembi Phiri	Royal AM	2005-06-03	20	173
8	Gugu Mahlangu	Bloemfontein Celtic	1993-10-22	32	171
9	Nokuthula Mab...	SuperSport United	1998-02-03	27	167
10	Khanyi Baloyi	Bloemfontein Celtic	1998-08-08	27	163
11	Zanele Phiri	TS Galaxy	1993-09-12	32	161
12	Kagiso Dlamini	Golden Arrows	1993-12-20	32	159
13	Thabo Phiri	Moroka Swallows	2003-06-11	22	156
14	Karabo Tshaba...	Stellenbosch FC	1991-02-18	34	155
15	Mona Muela	Moroka Swallows	2003-11-24	22	155

Query executed successfully. DESK